

Jason Tunstill

Software Engineer | C++ | Python | Qt | Ruby on Rails | Linux | Windows

Murfreesboro, TN • 615-295-5011 • tunstij@gmail.com

LinkedIn: [linkedin.com/in/jasontunstill](https://www.linkedin.com/in/jasontunstill) • GitHub: github.com/ironhead3829 • Portfolio: ironhead3829.github.io/bio

SUMMARY

- Software Engineer with 11+ years of experience building and supporting production software across desktop, backend, web, and engineering systems.
- Strong background in C++, Python, Qt, Ruby on Rails, Linux, REST APIs, hardware/software integration, CI/CD, and complex debugging.
- Experienced in modernizing legacy systems, designing maintainable software components and systems, optimizing application performance, and mentoring developers.
- Broad engineering background spanning application development, data acquisition, hardware integration, test systems, and end-to-end system troubleshooting.

TECHNICAL SKILLS

Languages: C++, C, Python, C#, Ruby, JavaScript, TypeScript, SQL, HTML/CSS, Bash, PowerShell, LabVIEW

Frameworks & Technologies: Qt, QML, PySide6, Ruby on Rails, Flask, FastAPI, SWIG, FLTK, jQuery, REST APIs, ZeroMQ

Systems & Integration: REST APIs, ZeroMQ, JSON, SWIG, DAQ hardware integration, FlexNet Publisher, client/server systems

Desktop & Systems Development: Cross-Platform Development, GUI Development, Multithreaded Programming, Client/Server Architecture, IPC, Hardware Integration, DAQ Systems, FlexNet Publisher

Build, DevOps & CI/CD: Git, GitHub, GitLab, CMake, Conan, Docker, GitLab CI, Azure DevOps, Azure Pipelines, Qt Installer Framework, Build Automation

Databases: PostgreSQL, MySQL, MariaDB

Operating Systems: Linux (Ubuntu, Debian, CentOS, Oracle Linux, RHEL), Windows 10/11, Windows Server

Software Engineering: Object-Oriented Programming/Design (OOP/OOD), Design Patterns, Debugging, Performance Optimization, Code Reviews, Refactoring, Technical Documentation, Agile/Scrum, Mentoring

AI-Assisted Development: GitHub Copilot, ChatGPT, AI-assisted debugging, code review, refactoring, and codebase analysis

EXPERIENCE

Software Engineer — Amentum, Tullahoma, TN | March 2026 – Present

- Redesigned a C#/NET API integrating LabVIEW applications with DevExpress XtraGrid, resolving DLL loading and cross-threading issues while significantly improving application stability.
- Increased real-time display performance from approximately 0.7 updates/sec to 10 updates/sec (15× improvement) by redesigning data exchange between LabVIEW and DevExpress XtraGrid.
- Develop and maintain LabVIEW and C# applications supporting real-time test and data acquisition systems, including controls/displays, Redis and T-SQL data interaction, and LabVIEW integration components.
- Design and implement new features for LabVIEW-based mission-critical test and data acquisition applications, enhancing usability and expanding operational capabilities.
- Collaborate with cross-functional engineering teams to modernize legacy LabVIEW applications through improved software architecture, UI responsiveness, and maintainability.

Software Developer — Apex Turbine (formerly Experimental Design & Analysis Solutions), Spring Hill, TN | January 2015 – January 2026

- Developed and enhanced cross-platform C++ engineering applications using Qt and FLTK, including data visualization, test configuration, limit-definition, and automated engineering-report generation tools.
- Modified and maintained C++ hardware integration plugins for National Instruments, VTI, ME-CALC, General Standards, and other DAQ systems, troubleshooting hardware/software interactions and integrating vendor driver APIs.
- Developed a QML desktop client that communicated with a Python Flask server through REST endpoints for application services and data exchange.
- Architected a cross-platform Python licensing daemon integrating modernized C++17 FlexNet Publisher wrappers through SWIG and a custom ZeroMQ REQ/REP JSON protocol; provided a common licensing API, CLI, and PySide6 GUI used by 6 desktop applications.
- Implemented and maintained the response side of the OpenCode offline activation/deactivation workflow in Ruby on Rails, managing license counts and expirations while supporting FlexNet Trusted Storage without requiring direct Internet access on licensed systems.
- Primary developer and maintainer of a legacy Ruby on Rails business platform, upgrading Rails 4.1 through 6.0 and maintaining internal CRM, inventory, quoting, timekeeping, licensing, and engineering workflows. Contributed to migration from MariaDB to PostgreSQL and from a CentOS VM deployment to Docker in Microsoft Azure.
- Built and enhanced internal timekeeping, CRM, contract management, inventory, quoting, and engineering workflows, including PDF/Excel quote generation, jQuery/AJAX interfaces, scheduled automation, and database/query optimization.
- Designed v1/v2 REST APIs and webhooks for bidirectional synchronization between customer support tickets in Rails and Azure DevOps work items, including ticket creation, status, priority, and comments.
- Created and maintained GitLab CI and Azure DevOps pipelines for Windows/Linux build, packaging, artifacts, installers, and releases; maintained QtIFW installers and troubleshoot build, release, and customer installation failures.

- Co-developed a Python/Type build CLI integrating CMake configuration, compilation, Windows code signing, install staging, Python wheel generation, packaging, and Azure Pipelines artifact creation.
- Performed complex debugging across C++, Python, and Ruby on Rails codebases; conducted merge-request reviews, mentored junior developers and interns, and helped onboard engineers to mature codebases.
- Worked directly with dozens of customer teams on requirements, bug reproduction, crash/log analysis, hardware and data troubleshooting, hotfixes, software deployment, and technical support.

Associate Engineer / Project Lead — QualTech NP (Curtiss-Wright), Huntsville, AL | July 2009 – January 2015

- Led multiple electrical penetration assembly (EPA) projects from planning through completion, coordinating engineering activities, schedules, documentation, and customer deliverables.
- Developed assembly procedures, manufacturing documentation, and acceptance test procedures supporting production and qualification of electrical penetration assemblies.
- Designed qualification test procedures and technical reports for research and development programs in accordance with IEEE 317 and applicable industry standards.
- Developed qualification test procedures and analyzed test results for LOCA, seismic, insulation resistance, helium leak, dielectric, and environmental testing in accordance with IEEE 317 requirements.
- Tracked project progress, managed engineering documentation, and coordinated cross-functional teams to meet technical, quality, and schedule requirements.
- Prepared technical reports, analyzed test data, and documented qualification results supporting product certification and customer acceptance.

EDUCATION

Bachelor of Science, Computer Engineering — University of Alabama in Huntsville | May 2014